

Algorithmic Aspects of Independent Roman Domination

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Abstract. For a graph $G = (V, E)$, an independent Roman domination function (IRDF) is a function $f : V(G) \rightarrow \{0, 1, 2\}$ such that every vertex v with $f(v) = 0$ is adjacent to a vertex u with $f(u) = 2$, where the set of vertices assigned positive values forms an independent set. The independent Roman domination number, denoted by $\gamma_{iR}(G)$, is the minimum weight over all IRDFs. In this work, we present an exact polynomial-time algorithm running in $O(n^4)$ time to compute $\gamma_{iR}(G)$ for asteroidal triple-free (AT-free) graphs. Additionally, since the Minimum Independent Roman Domination (MIN-IRD) problem is NP-complete on general graphs, we provide a fixed-parameter tractable (FPT) algorithm parameterized by the treewidth of the input graph, running in $O(5^{\text{tw}(G)} \cdot n)$ time.

1 Introduction and Invariant Background

Domination in graphs is one of the most well-studied problems in graph theory and has been investigated for more than fifty years. A dominating set of a graph $G = (V, E)$ is a subset $D \subseteq V(G)$ such that every vertex in $V(G) - D$ is adjacent to at least one vertex in D . Roman domination is a variation of domination introduced by Ian Stewart in his article “Defend the Roman Empire!” [16]. It was inspired by the defense strategy of the Roman Empire, where two legions were placed at selected locations such that any unprotected region could be defended by relocating a nearby legion when needed. A Roman dominating function on a graph G is a function $f : V(G) \rightarrow \{0, 1, 2\}$ such that every vertex assigned 0 has at least one adjacent vertex assigned 2. Due to applications of Roman domination in areas such as network security, communication networks, and resource allocation, many variants of Roman domination have been studied. One important variation is independent Roman domination [6]. In this variation, the set of vertices having positive assignment (that is, vertices assigned 1 or 2) must form an independent set. The weight of an independent Roman dominating function is $w(f) = \sum_{v \in V(G)} f(v)$. The minimum weight among all independent Roman dominating functions on G is called the independent Roman domination number of G and is denoted by $\gamma_{iR}(G)$. More details about Roman domination and its variants can be found in [5, 6, 14].

MINIMUM INDEPENDENT ROMAN DOMINATION problem (MIN-IRD)

Instance: A graph $G = (V, E)$.

Solution: $\gamma_{iR}(G)$.

The decision version of the problem is denoted as the DECIDE-IRD problem. MIN-IRD problem was introduced as a variant of the Roman domination problem by Cockayne et al. [6]. Targhi et al. [17] established some relations between independent Roman domination, Roman domination and found some bounds. Duan et al. [9] showed that the DECIDE-IRD problem is NP-complete for chordal bipartite graphs. In the same paper, they have also proposed a linear-time algorithm for the MIN-IRD problem on trees.

Since the DECIDE-IRD problem is NP-complete for bipartite graphs [6], it is natural to study it from the perspective of parameterized complexity. For the Roman domination problem, Fernau et al. developed a parameterized algorithm for general graphs [10]. In this work, we propose a parameterized algorithm for the MIN-IRD problem and a polynomial-time algorithm for AT-free graphs.

2 Exact Polynomial-Time Bounds on AT-free Graphs

An asteroidal triple (AT) consists of three independent vertices such that a path exists between any two that avoids the closed neighborhood of the third. Graphs that do not contain such triplets form the family of AT-free graphs [13], which is a superclass of interval graphs, permutation graphs, and cocomparability graphs. Due to structural properties and decomposition, some variants of domination problems can be solved efficiently on AT-free graphs [1, 8].

Our exact algorithm uses a structural partition of an AT-free graph [4]. It decomposes the graph into smaller AT-free subgraphs. By placing a weight of 2 on a choice vertex v , the global problem decomposes into independent sub-problems on the connected components of $G \setminus N[v]$. We introduce a systematic sub-component classification framework:

- **Type 1 Components:** Connected sub-components C' satisfying $N[V(C')] \cap N[v] \neq \emptyset$. We prove these components are contained within the structural interval $I_{u,v}$ between nonadjacent endpoints.
- **Type 2 Components:** Connected sub-components where $N[V(C')] \cap N[v] = \emptyset$. These are shown to isolate cleanly as maximal connected components in $G \setminus N[v]$.

By utilizing interval structural properties defined by Broersma et al. [4], we decompose the AT-free graphs into smaller components and intervals. This clear decomposition gives an idea of a recursive dynamic programming algorithm. By sorting the components and intervals according to their size, the algorithm computes the recurrence values in a bottom-up manner and computes the exact value of $\gamma_{iR}(G)$ in $O(n^4)$ time. This matches efficient bounds known for related problems on similar graph structures [12, 15, 7].

3 Fixed-Parameter Tractability via Treewidth

We also investigate the MIN-IRD problem from a parameterized perspective. We develop a dynamic programming algorithm based on a nice tree decomposition [2] of the input graph, parameterized by its treewidth $tw(G)$. Several parameterized algorithms using treewidth as a parameter can be found in [11, 2, 3]. The algorithm processes the decomposition in a bottom-up manner using four labels, $\{0, 0^*, 1, 2\}$. Here, the labels 0, 1, and 2 correspond to vertices assigned the values 0, 1, and 2, respectively, and represent their status with respect to domination and independence constraints. The label 0^* is used to record temporarily undominated vertices whose domination may be established during subsequent computation. The algorithm processes the leaf, introduce, forget, and join nodes of the nice tree decomposition while maintaining all feasible partial solutions to compute the minimum independent Roman domination number. The resulting algorithm runs in $O(5^{tw(G)}n)$ time, establishing that MIN-IRD is fixed-parameter tractable when parameterized by the treewidth of the input graph.

4 Conclusion

In this paper, we have proposed an algorithm that computes the independent Roman domination number of an AT-free graph in $O(n^4)$ time complexity. We have also proposed a fixed parameterized algorithm with parameter as treewidth to solve this problem with time complexity $O(5^{tw(G)}n)$.

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